

SingleRAN

Software Management Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.0 01 (2025-09-16)

This issue does not include any changes.

1.2 SRAN22.0 Draft A (2025-06-30)

This issue does not introduce any changes to SRAN21.1 01 (2025-03-10).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
SRAN	MRFD-210402	BTS/NodeB Software Management	3.1 Overview
LTE FDD	MRFD-210401	BSC/RNC Software Management	
LTE FDD	LBFD-004004	Software Version Upgrade Management	
LTE FDD	LBFD-004005	Hot Patch Management	
NB-IoT	MLBFD-12000404	Software Version Upgrade Management	
NB-IoT	MLBFD-12000405	Hot Patch Management	
LTE TDD	TDLBFD-004004	Software Version Upgrade Management	
LTE TDD	TDLBFD-004005	Hot Patch Management	
NR	FBFD-010025	Basic O&M Package (specifically, the software version upgrade management and hot patch management functions)	

3 Software Management

3.1 Overview

Introduction

Software management implements upgrades, checks, and rollbacks of software versions and patches through the MAE, local maintenance terminal (LMT), or the tool in the version package.

Related Concepts

Concept	Description
NE software	Software running on NEs for implementing specific functions. Each NE software has a unique version number.
Baseline patch	Full SPC software version of a certain release, applicable only to base station controllers
Software patch	Software patches are classified into hot patches and cold patches. Each software patch has its own version number and can be used to fix certain defects of the NE software.
Cold patch	A cold patch upgrade takes effect only after some boards are or the NE is restarted. Restarting boards will interrupt some services, and restarting the NE will interrupt its services for a short period of time. The cold patch of the base station is the full SPC software version of a certain release.
Hot patch	NEs do not need to be restarted after hot patches are installed, which does not affect ongoing NE services.

Concept	Description
MAE-Access mediation	It helps the MAE-Access to adapt to NEs of different types. It is recommended that you install the MAE-Access mediation before upgrading NEs.
Upgrade risk check package	The upgrade risk check package is used to verify the upgrade feasibility and check for upgrade risks.
Upgrade path	Version path from the source version to the target version in an upgrade
MAE-Deployment component package	The MAE-Deployment is the data configuration management system for Huawei radio access networks (RANs), such as GBSS, WRAN, eRAN, SRAN, and 5G RAN. The MAE-Deployment is integrated into the MAE as a configuration tool component. It is at the element management system (EMS) layer and enables users to configure NEs in a unified manner.
Extended patch	It is used for the software upgrades of the following: (1) EOS boards (to the last supported version); (2) new boards (developed in the new version but used in the current version in advance); (3) some third-party components. The installation of the extended patch affects only the board to be upgraded and does not affect the services of other boards.

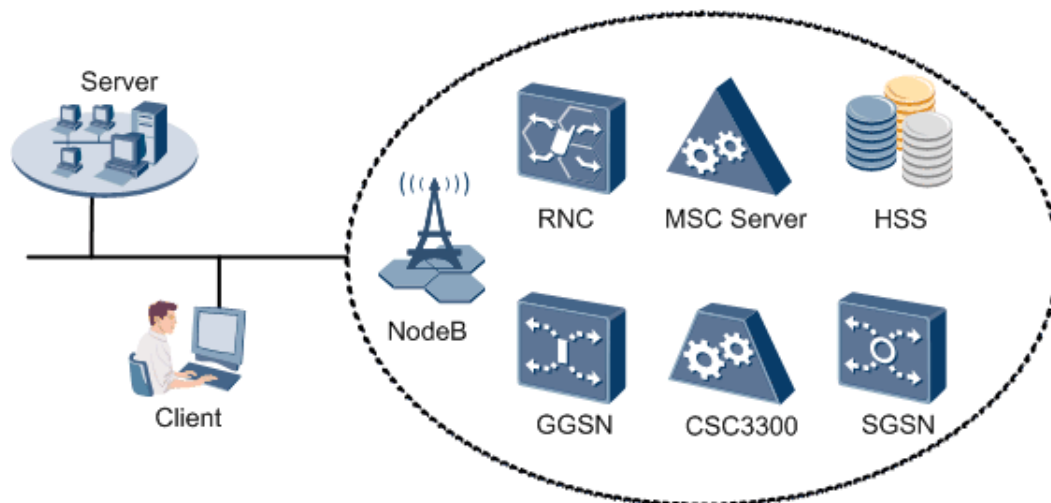
Software Management Modes

To implement operation and maintenance (O&M) in different scenarios, Huawei provides two software management modes: managing software remotely and centrally and managing software locally.

When managing software remotely and centrally, users perform operations on a group of NEs using software management functions provided by the MAE, as shown in [Figure 3-1](#). The operations include distributing and backing up version files as well as loading and activating software.

This mode is efficient and applicable to the normal network O&M. Users can perform batch operations through the OSS, reducing O&M costs. However, if the operation and maintenance channel (OMCH) fails to be established between an NE and the MAE, this mode does not work.

Figure 3-1 Managing software remotely and centrally

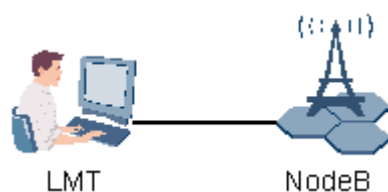


When managing software of an NE locally, users use the LMT connected to the NE directly to distribute and back up version files as well as load and activate software, as shown in [Figure 3-2](#).

This mode requires site visits, which are inefficient and incur high expenses. Therefore, this mode is recommended only in the following scenarios:

- The OMCH between the NE and the MAE has not been established during new NE deployment.
- Network faults occur or the OMCH between the NE and the MAE is disconnected.
- The MAE does not support software management for the NE.

Figure 3-2 Managing software locally



3.2 Upgrade Policy

3.2.1 Network Upgrade Policy

You are advised to perform upgrades in the following sequence: MAE-Access/MAE-Deployment > BSC > USU > base station.

If there are USUs deployed on a RAN, it is required that all NEs under these USUs (including the USUs and base stations) be upgraded in the same batch. If the USUs and base stations cannot be upgraded in the same batch, upgrade all the USUs in a batch and all the base stations within 24 hours after the USU upgrade.

3.2.2 Base Station Upgrade Policy

Two upgrade methods are available:

- By base station
The upgrade is performed on the MAE by base station (for example, a single-mode base station, a co-MPT base station, or a separate-MPT base station). In a multimode base station, main control boards of different RATs are upgraded simultaneously and therefore services of all RATs are interrupted simultaneously. This reduces service interruptions and shortens the interruption duration.
- By NE
The upgrade is performed on the MAE by NE (for example, a single-mode base station, a co-MPT base station, or an NE of a separate-MPT base station). Services of different NEs in a separate-MPT base station are interrupted separately.

Recommended Upgrade Policy

Table 3-1 lists the upgrade policy recommended in each scenario.

Table 3-1 Recommended upgrade policies

Scenario	Recommended Upgrade Policy
Separate-MPT multimode base station	• If simultaneous interruption of services in all modes is acceptable, an upgrade by base station is recommended. • Otherwise, an upgrade by NE is recommended.
Single-mode base station or co-MPT multimode base station	An upgrade by base station is recommended for a co-MPT multimode base station, which also supports an upgrade by NE.

NE Sequence During an Upgrade by NE

During an upgrade by NE, if the base station can be directly upgraded to the target version, there is no requirement on the NE sequence. For a co-RRU base station or a base station implementing common transmission, you are advised to upgrade all NEs to the target version in the same upgrade time window.

If the base station needs to be upgraded to an interim version and then to the target version, first upgrade the NE operating in the latest version. If the versions of all modes are the same, first upgrade the NE of the most recent mode. It is recommended that you upgrade all the NEs to the interim version before upgrading them to the target version.

3.2.3 Hot Patch Upgrade Policy

A hot patch of a base station controller can be installed based on the original baseline patch, cold patch, or hot patch.

Hot patches of a base station strictly match patch versions. That is, one hot patch corresponds to only one cold patch. Assume that a hot patch exists in the base station and you want to install a new hot patch. For the GBTS, you need to uninstall the existing hot patch and then install the new hot patch; for other RATs, you can directly install the new hot patch.

3.3 NE Upgrade Mode (Recommended)

Table 3-2 lists the recommended modes for upgrading base station controllers, base stations, and USUs. For detailed upgrade operations, see the NE upgrade guide.

Table 3-2 Recommended upgrade modes

NE	Recommended Upgrade Mode	Description
Base station or USU	MAE software management expert (SME)	The MAE SME allows you to upgrade NEs by project or by batch. It supports batch upgrade planning and management and pre-upgrade risk check, significantly improving upgrade efficiency.
Base station controller	Upgrade tool in the base station controller software package	The upgrade tool in the base station controller software package is used for NE version upgrades.

MAE SME

The MAE SME provides an end-to-end upgrade solution. To meet engineers' O&M requirements, the end-to-end upgrade solution involves phases such as:

- Network-wide upgrade planning
- Pre-upgrade risk check
- Upgrade execution
- Comparison of alarms, KPIs, and cell status before and after the upgrade

The SME has the following characteristics:

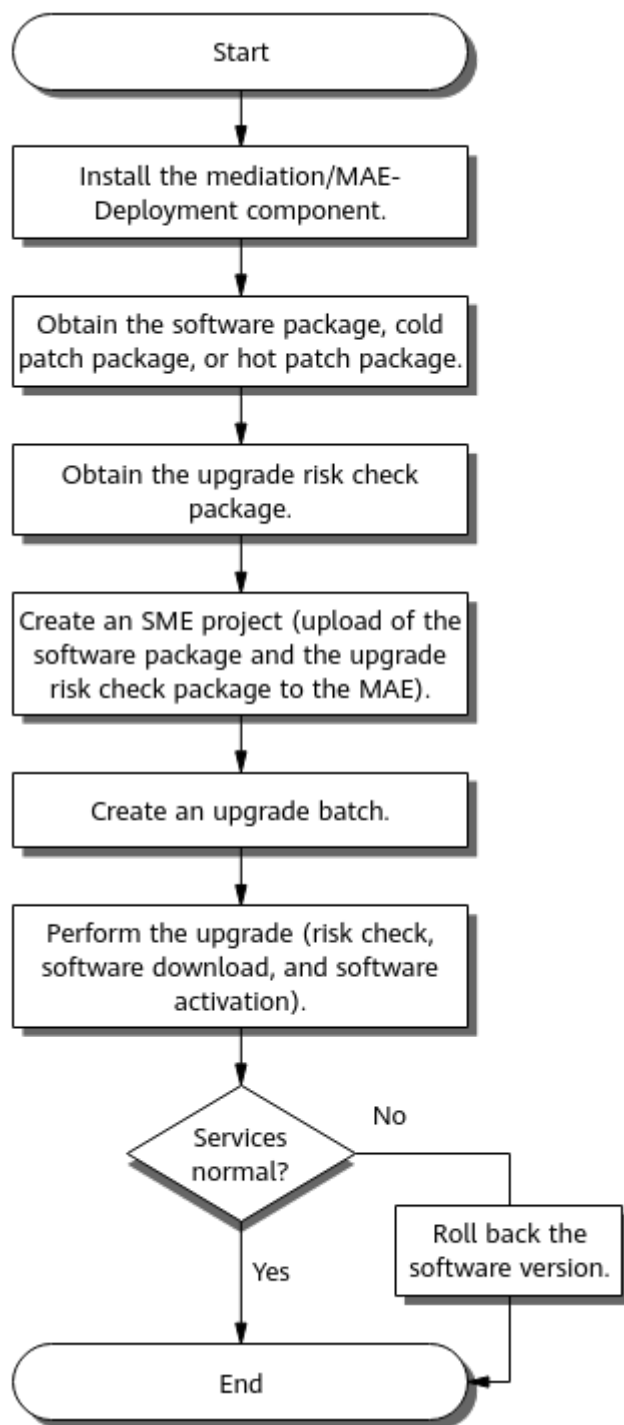
- Network-wide upgrade planning
Users can view base station distribution under base station controllers or in subnets, plan upgrades across the network by area, and create batch upgrade projects to upgrade NEs across the network.
- Integrated various phases in the upgrade procedure
The phases include pre-upgrade risk check, upgrade execution, and comparison of cell status and alarms before and after the upgrade. Instead of

understanding the entire upgrade process, users only need to perform step-by-step operations according to the instructions provided in each upgrade phase.

- Integrated pre-upgrade risk check and comparison of cell status and alarms
On the MAE, users can perform the pre-upgrade risk check by using the upgrade risk check packages of NEs. In the pre-upgrade risk check phase, the SME checks the version mapping and the upgrade path, and checks for upgrade risks of each version. Comparison of cell status and alarms before and after the upgrade helps users learn the differences of cell status and alarms before and after the upgrade. These functions automate manual operations and improve upgrade efficiency.
- Unified entry for upgrades of GBTs, eGBTs, NodeBs, eNodeBs, gNodeBs, and MBTs
GBTs, eGBTs, NodeBs, eNodeBs, gNodeBs, and MBTs can be upgraded using the same procedure, which improves the O&M experience.

Figure 3-3 shows the procedure for upgrading NEs through the MAE SME.

Figure 3-3 Upgrading NEs through the MAE SME

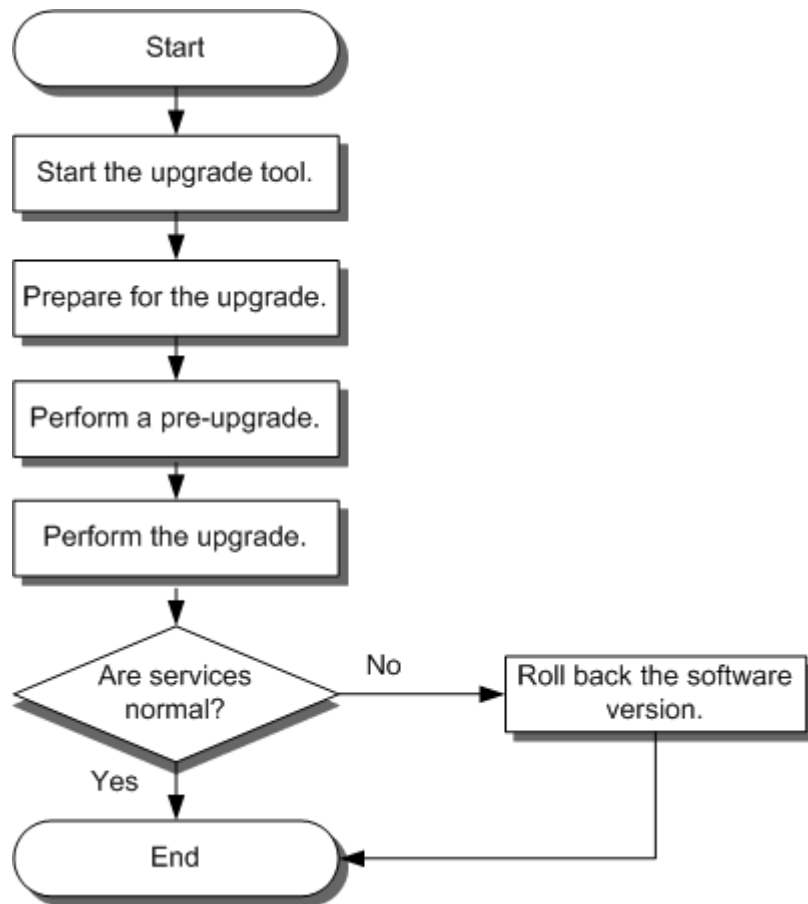


Upgrade Tool for Base Station Controllers

The upgrade tool in the controller software package is used for version and patch upgrades of base station controllers.

Figure 3-4 shows the procedure for upgrading a base station controller through the upgrade tool.

Figure 3-4 Upgrading a base station controller through the upgrade tool



4 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

5 Reference Documents

None.